1 Rings and Ideals

Exercise 1.1. Let x be a nilpotent element of a ring A. Show that $1+x$ is a unit of A. Deduce that the sum of a nilpotent element and a unit is a unit.

Proof: Idea: Think about Jordan blocks of matrices. Let $n \in \mathbb{N}$ be the smallest integer such that $x^n = 0$. Let

$$w = \sum_{k=0}^{n-1} (-1)^k x^k.$$

Then

$$\begin{aligned} w(1+x) &= \left(1 + \sum_{k=1}^{n-1} (-1)^k x^k\right) + \sum_{k=1}^{n} (-1)^{k-1} x^k \\ &= 1 + \sum_{k=1}^{n-1} x^k\big((-1)^k + (-1)^{k-1}\big) = 1 \end{aligned}$$

Suppose $a \in R$ is a unit. Then let

$$(a+x)^{-1} = \frac{1}{a} \sum_{k=0}^{n-1} (-1)^k \left(\frac{x}{a}\right)^k$$

□

Exercise 1.2. Let A be a ring and let $A[x]$ be the ring of polynomials in an indeterminate x, with coefficients in A. Let $f = a_0 + a_1 x + ... + a_n x^n \in A[x]$. Prove that

i) f is a unit in $A[x] \Leftrightarrow a_0$ is a unit in A and $a_1, ..., a_n$ are nilpotent. [If $b_0 + b_1 x + ... + b_m x^m$ is the inverse of f, prove by induction on r that $a_n^{r+1} b_{m-r} = 0$. Hence show that a_n is nilpotent, and then use Ex. 1.]

ii) f is nilpotent $\Leftrightarrow a_0, a_1, ..., a_n$ are nilpotent.

iii) f is a zero-divisor $\Leftrightarrow$ there exists $a \neq 0$ in A such that $af = 0$. [Choose a polynomial $g = b_0 + b_1 x + ... + b_m x^m$ of least degree m such that $fg = 0$. Then $a_n b_m = 0$, hence $a_n g = 0$ (because $a_n g$ annihilates f and has degree $< m$). Now show by induction that $a_{n-r} g = 0$ $(0 \leq r \leq n)$.]

iv) f is said to be **primitive** if $(a_0, a_1, ..., a_n) = (1)$. Prove that if $f, g \in A[x]$, then fg is primitive $\Leftrightarrow f$ and g are primitive.

Proof:

i) $\Rightarrow$: This direction requires two steps of induction. We will first induct on the degree and then internally we will do a secondary induction to show that the last coefficient is nilpotent. For all degree 0 units in $A[x]$ the statement is trivially true. Suppose for the sake of induction, that for all degree $k \leq n$ polynomials that are units that the constant term is a unit in A and the higher coefficients are nilpotent. Suppose f is a degree $n+1$ polynomial

$$f = \sum_{k=0}^{n+1} a_k x^k,$$

such that f is a unit in $A[x]$. Let

$$g = \sum_{l=0}^{m} b_l x^l$$

be the inverse of f. Then we have that

$$\begin{aligned} fg &= \sum_{k=0}^{n+1} \left(\sum_{l=0}^{m} a_k b_l \right) x^{k+l} \\ &= \sum_{k=0}^{n+m+1} \left(\sum_{l=\max(0,k-m)}^{\min(n+1,k)} a_l b_{k-l} \right) x^k \\ &= 1 \end{aligned}$$

Thus we know that $a_{n+1} b_{n+1} = 0$. Assume for the sake of induction that

$$a_{n+1}^{s+1} b_{m-s} = 0$$

for all $s \leq r$. Then examine the coefficient of x^{n+m-r} in fg we have the following sum

$$\begin{aligned} 0 &= \sum_{l=\max(0,n-r)}^{\min(n+1,n+m-r)} a_l b_{(n+m-r)-l} \\ &= \sum_{l=\max(0,n-r)}^{n+1} a_l b_{(n+m)-r-l} \\ &= a_{n+1}^{r+1} \sum_{l=\max(0,n-r)}^{n} a_l b_{(n+m)-r-l} \\ &= a_{n+1}^{r+2} b_{(n+m)-n-1-r} = a_{n+1}^{r+2} b_{m-(r-1)}. \end{aligned}$$

Thus by induction we know that $a_{n+1}^m b_0 = 0$. Since b_0 must be a unit, $a_{n+1}^m = 0$. Since f is a unit, $f - a_{n+1} x^{n+1}$ is a unit and by the inductive hypothesis, a_1, ..., a_n are nilpotent.

$\Longleftarrow$: Immediate using exercise 1.

ii) $\Rightarrow$: Suppose $f = \sum_{k=0}^{n} a_k x^k$ is nilpotent. Then $f + 1$ is unit. But then we know that $a_1, ..., a_n$ are nilpotent. To get that a_0 is nilpotent, just look at the constant term in $f^k = 0$ which is a_0^k, it must be zero and thus $a_0^k = 0$.

$\Longleftarrow$: We will prove a general statement. Suppose $a, b \in R$ are nilpotent. Then $a + b$ is nilpotent. Suppose $a^n = 0$ and $b^m = 0$. Then

$$\begin{aligned} (a+b)^{n+m} &= \sum_{k=0}^{n+m} \binom{n+m}{k} a^k b^{n+m-k} \\ &= \sum_{k=0}^{n} \binom{n+m}{k} a^k b^{m+(n-k)} + \sum_{k=n+1}^{n+m} \binom{n+m}{k} a^k b^{m+n-k} \\ &= 0 \end{aligned}$$

Thus $a + b$ is nilpotent. In fact, the set of nilpotent elements form a ideal. This means that f is nilpotent since $a_k x^k$ is nilpotent for all $k \in \{0, ..., n\}$.

iii) Only one direction is interesting. We first prove the claim that if $f = [a_0, ..., a_m] \in A[x]$ is a zero-divisor, then its corresponding g (in the context of this problem) has the property that $a_k g = 0$ for all k. We will do this by induction on the degree of f. The claim is trivially true

for degree 0 zero-divisor. Suppose the claim is true for all zero-divisor divisors of degree less than or equal to n. Let $f = [a_0, ..., a_{n+1}]$ be a degree $n+1$ zero-divisor and let $g = [b_0, ..., b_m]$ be its corresponding zero-divisor (of minimal degree). The x^{m+n+1} coefficient of fg is $a_{n+1}b_m = 0$. Then $a_{n+1}g$ has smaller degree than g and $a_{n+1}gf = 0 \Rightarrow a_{n+1}g = 0$ by minimality of g. Now consider $f' = f - a_{n+1}x^n$ is a polynomial of degree at most n. Furthermore, $f'g = fg - a_{n+1}g = 0$. By the inductive hypothesis, each coefficent of the polynomial must multiply with g to equal 0, the coefficients of f' are $a_0, ..., a_n$. Thus $b_m a_k = 0$. Then $b_m f = 0$.

iv) Going forward doesn't seem too bad, basically you show that the ideal generated by the coefficients of fg is contained in the ideal $(a_1, ..., a_n)$ (and equivalently $(b_1, ..., b_n)$). You basically just use the coefficients of fg are polynomial in the coefficents of f and g themselves with the coefficients lying in A, hence ideals are closed under addition and multiplication by elements in A. Suppose f and g are primitive. Assume the contrary, that fg is not primitive, let $c(fg)$ denote the ideal generated by the coefficients of fg. Thus there exists a maximal ideal $\mathfrak{m} \supset c(fg)$. $A/\mathfrak{m}$ is a field. Consider $f, g \in A/\mathfrak{m}[x]$ (image under quotient map) are not zero divisors since they are nonzero and $(A/\mathfrak{m})[m]$ is a integral domain. The image of fg under the quotient map is 0 though which is a contradiction. Hence fg must be primitive.

□

Exercise 1.3. Generalize the results of Exercise 2 to a polynomial ring $A[x_1, ..., x_r]$ in several indeterminates.

Proof: Note the following, there is a isomorphism between $A[x_1, ..., x_r] \cong (A[x_1, ..., x_{r-1}])[y]$. We will basically exploit this isomorphism to run a inductive argument on the number of indeterminates. Suppose the statements are true for polynomials in $A[x_1, ..., x_k]$. Let $f \in A[x_1, ..., x_{k+1}]$, f can be viewed as a element of $A[x_1, ..., x_k][x_k]$.

$$f = \sum_{i=0}^{n} P_i x_r^i$$
$$P_i \in A[x_1, ..., x_k].$$

Apply the previous exercise to get statements about the coefficients P_k and then use inductive hypothesis. □

Exercise 1.4. In the ring $A[x]$, the Jacobson radical is equal to the nilradical.

Proof: We want to show that $\mathfrak{N} \subseteq \mathfrak{R}$ (nilradical is contained in Jacobson radical) and $\mathfrak{R} \subseteq \mathfrak{N}$. All maximal ideals are prime, hence the intersection of all prime ideals $\mathfrak{N}$ must lie in the intersection of all maximal ideals $\mathfrak{R}$. So the nontrivial direction is showing that

$$\mathfrak{R} \subseteq \mathfrak{N}.$$

So now we need to show that if $f \in \mathfrak{R}$, ie in the intersection of all maximal ideals of $A[x]$, it is also in $f \in \mathfrak{N}$, or in the intersection of all prime ideals of $A[x]$. By Prop 1.9 (Atiyah-Macdonald), we know that

$$f \in \mathfrak{R} \Leftrightarrow 1 - fg \text{ unit}$$

for any choice of g. Thus we know that $1 - fx$ is unit. Hence all the coefficients of f must be nilpotent. Hence f must be nilpotent, hence $f \in \mathfrak{N}$. □

Exercise 1.5. Let A be a ring and let $A[[x]]$ be the ring of formal power series $f = \sum_{n=0}^{\infty} a_n x^n$ with coefficients in A. Show that

i) f is a unit in $A[[x]] \Leftrightarrow a_0$ is a unit in A.
ii) If f is nilpotent, then a_n is nilpotent for all $n \geq 0$. Is the converse true? (See Chapter 7, Exercise 2.)
iii) f belongs to the Jacobson radical of $A[[x]] \Leftrightarrow a_0$ belongs to the Jacobson radical of A.
iv) The contraction of a maximal ideal m of $A[[x]]$ is a maximal ideal of A, and m is generated by m^e and x.
v) Every prime ideal of A is the contraction of a prime ideal of $A[[x]]$.

Proof: **(a)**: Refer to homework 2 where a more general arguement is given for $A[[x_1, ..., x_n]]$.

(b): Suppose $f = [a_i]_{i=0}^{\infty}$ is nilpotent. Consider $g_k = [a_0, ..., a_k]$, we will show that g_k is nilpotent for all $k \in \mathbb{N}$ hence $a_0, ..., a_k$ are nilpotent. Let

$$0 = f^n = \sum_{k=0}^{\infty} b_k x^k$$

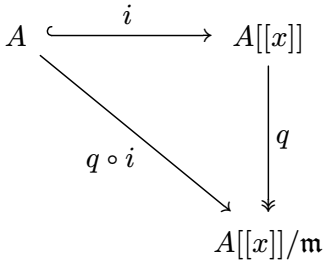
not that b_k for $k \in \{0, ..., nm\}$ are the coefficients of g_m^n. Thus since $b_k = 0 \Leftrightarrow g_m^n = 0 \Rightarrow g_m$ nilpotent. I am not able to say definitively whether or not a_n being nilpotent implies that f is nilpotent. But we know all finite sums of f are nilpotent. But I am not able to get anything stronger.

(c): Suppose $f = \left[\{a_n\}_{n=0}^{\infty}\right] \in \mathfrak{R}$ of $A[[x]]$. Thus for all $b \in A$, we know that $1 - fb$ is unit. Hence $1 - a_0 b$ is unit for any choice of $b \in A$. Hence by Prop. 1.6 (Atiyah-Macdonald) we know that $a_0 \in \mathfrak{R}$ of A. Suppose $a_0 \in \mathfrak{R}$ of A, for all $g \in A[[x]]$

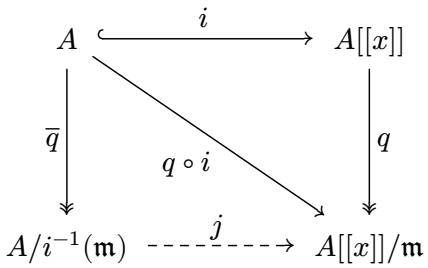
$$1 - fg \text{ unit} \Leftrightarrow 1 - g_0 a_0 \text{ unit}$$

and $1 - g_0 a_0$ is unit because $a_0 \in \mathfrak{R}$ of A. Hence by 1.9 again, $f \in \mathfrak{R}$ of $A[[x]]$.

(d): Note the following commutative diagram.



Note that $\ker(q \circ i) = i^{-1}(\mathfrak{m})$ hence $q \circ i$ factors through $\bar{i} : A \to A/i^{-1}(\mathfrak{m})$ and we get the following diagram.



Note that j is a injection, bc $j(k + i^{-1}(\mathfrak{m})) = i(j) + \mathfrak{m} = \mathfrak{m} \Leftrightarrow i(j) \in \mathfrak{m} \Rightarrow j \in i^{-1}(\mathfrak{m})$. Thus $i^{-1}(\mathfrak{m})$ must be at least a prime ideal of A because $A/i^{-1}(\mathfrak{m})$ injects into a field and hence must at least be a domain. We know that $0 \in \mathfrak{R}$ trivially. Then for any $[a_n]_{n=1}^{\infty} \in A[[x]]$, power series with constant term 0, $[a_n]_{n=1}^{\infty} \in \mathfrak{R}$. Consider $a + i^{-1}(\mathfrak{m}) \neq 0$.

$$a + i^{-1}(\mathfrak{m}) = a + \mathfrak{m}$$

through j. $\exists f = [f_k]_{k=0}^{\infty} \in A[[x]]$ such that

$$af = 1 \in A[[x]]/\mathfrak{m}.$$

$$\sum_{k=1}^{\infty} f_k x^k \in \mathfrak{R} \subseteq \mathfrak{m}.$$

Thus

$$af_0 = a\left(f - \sum_{k=1}^{\infty} f_k x^k\right) = 1 + a\sum_{k=1}^{\infty} f_k x^k = 1 \in A[[x]]/\mathfrak{m} \Rightarrow af_0 = 1 \in A/i^{-1}(\mathfrak{m})$$

$\forall g = [g_k]_{k=0}^{\infty} \in \mathfrak{m}$ we want to show that $g_k \in \mathfrak{m}^c$ for all k.

i) Case 1 $k = 0$: $[g_n]_{n=1}^{\infty} \in \mathfrak{R} \subset \mathfrak{m} \Rightarrow g - [g_n]_{n=1}^{\infty} = g_0 \in \mathfrak{m} \Rightarrow g_0 \in \mathfrak{m}^c$.

ii) Case 2 $k > 0$: Assume the contrary that $g_k \notin \mathfrak{m}^c$. By a similar trick as above and using the fact $g_0 \in \mathfrak{m}^c$ we can isolate the $g_k x^k$ term, $g_k x^k \in \mathfrak{m}$. Thus $(\mathfrak{m}^c, g_0)(x) \subseteq \mathfrak{m}$. $(\mathfrak{m}^c, g_0) = A$, my maximality. Thus $1 \in (\mathfrak{m}^c, g_0) \subset \mathfrak{m} \Rightarrow \mathfrak{m} = A[[x]]$, contradicting maximality of $\mathfrak{m}$.

□

Exercise 1.6. A ring A is such that every ideal not contained in the nilradical contains a nonzero idempotent (that is, an element e such that $e^2 = e \neq 0$). Prove that the nilradical and Jacobson radical of A are equal.

Proof: It suffices to show that

$$\mathfrak{R} \subseteq \mathfrak{N}.$$

Assume the contrary, that $\mathfrak{R} \nsubseteq \mathfrak{N}$. Thus $\exists e \in \mathfrak{R}$ such that $e \neq 0$ and $e^2 = e$. By Prop 1.9, $1 - e^2 = 1 - e$ is a unit. Let $f = (1 - e)^{-1}$. Note the following computation,

$$e = e(1 - e)f = (e - e^2)f = (e - e)f = 0.$$

This is a contradiction. Hence $\mathfrak{R} \subseteq \mathfrak{N}$ and $\mathfrak{R} = \mathfrak{N}$. □

Exercise 1.7. Let A be a ring in which every element x satisfies $x^n = x$ for some $n > 1$ (depending on x). Show that every prime ideal in A is maximal.

Proof: The problem is equivalent to showing that $A/\mathfrak{p}$ is a field for any prime ideal $\mathfrak{p}$. Fix $\mathfrak{p} \in \operatorname{Spec}(A)$. We want to show that $A/\mathfrak{p}$ is a field.

$$x^n - x = 0 = x(x^{n-1} - 1).$$

Thus in $A/\mathfrak{p}$ which is a domain, either $x = 0$ or $x^{n-1} - 1 = 0$. Thus if $x \neq 0$, then $x^{n-1} = 1$. Thus $x^{n-2}x = 1$ and x is unit. Thus every nonzero element in $A/\mathfrak{p}$ is unit and $A/\mathfrak{p}$ is a field and $\mathfrak{p}$ is maximal. □

Exercise 1.8. Let A be a ring $\neq 0$. Show that the set of prime ideals of A has minimal elements with respect to inclusion.

Proof: Let $\operatorname{Spec}(A)$ denote the set of prime ideals of A, and use $\supseteq$ as the comparison operator on the set. Let C be a arbitrary chain in $\operatorname{Spec}(A)$. Consider

$$I = \bigcap_{\mathfrak{p} \in C} \mathfrak{p},$$

which is an ideal of A. We want to show I is prime. For $\mathfrak{p} \in C$, let $f_{\mathfrak{p}} : A \to A/\mathfrak{p}$ be the standard quotient. Note that f factors through $\tilde{f}_{\mathfrak{p}} : A/I \to A/\mathfrak{p}$. Let $x, y \in A/I$ be elements where $xy = 0$ then $0 = \tilde{f}_{\mathfrak{p}}(xy) = \tilde{f}_{\mathfrak{p}}(x)\tilde{f}_{\mathfrak{p}}(y) \Rightarrow \tilde{f}_{\mathfrak{p}}(x) = 0$ or $\tilde{f}_{\mathfrak{p}}(y) = 0$. Meaning $\forall \mathfrak{p} \in C$, $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. Note the following, if $\mathfrak{p} \not\ni x \Rightarrow \mathfrak{q} \not\ni x \wedge I \not\ni x$ for all $\mathfrak{q} \in C$ where $\mathfrak{p} \supseteq \mathfrak{q}$.

i) $\forall \mathfrak{p} \in C$ suppose there $\exists z \in \{x, y\}$. Such that $z \in \mathfrak{p}$, thus $z \in I$ and either $x = 0$ or $y = 0$ in A/I.

ii) $\exists \mathfrak{p} \in C$ where $x \in \mathfrak{p} \in C$ but $x \notin \mathfrak{q} \in C$ ($y \in \mathfrak{q}$). If $\mathfrak{p} \supset \mathfrak{q} \Rightarrow y \in \mathfrak{p}$ and $y \in I \Rightarrow y = 0 \in A/I$. If $\mathfrak{q} \supseteq \mathfrak{p}$, this is not possible.

Thus A/I is a domain. By Zorn's Lemma, exists minimal elements in $\operatorname{Spec}(A)$. □

Exercise 1.9. Let a be an ideal $\neq (1)$ in a ring A. Show that $a = r(a) \Leftrightarrow a$ is an intersection of prime ideals.

Proof: $\Rightarrow$: Suppose $a = r(a)$. Consider $q : A \to A/a$, note that $a = r(a) = q^{-1}(\mathfrak{N}(A/a))$. Thus

$$a = q^{-1}\left(\bigcap_{p \in \operatorname{Spec}(A/a)} \mathfrak{p}\right) = \bigcap_{\mathfrak{p} \in \operatorname{Spec}(A/a)} q^{-1}(\mathfrak{p}) = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec}(A) \\ a \subseteq \mathfrak{p}}} \mathfrak{p}$$

$\Leftarrow$: Suppose

$$a = \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec}(A) \\ a \subseteq \mathfrak{p}}} \mathfrak{p}.$$

Let $q : A \to A/a$. Thus

$$
\begin{aligned}
a &= \bigcap_{\mathfrak{p} \in \operatorname{Spec}(A/a)} q^{-1}(\mathfrak{p}) \\
&= q^{-1}\left(\bigcap_{\mathfrak{p} \in \operatorname{Spec}(A/a)} \mathfrak{p} \right) \\
&= q^{-1}(\mathfrak{N}(A/a)) \\
&= r(a)
\end{aligned}
$$

□

Exercise 1.10. Let A be a ring, $\mathfrak{N}$ its nilradical. Show that the following are equivalent:

i) A has exactly one prime ideal;
ii) every element of A is either a unit or nilpotent;
iii) $A/\mathfrak{N}$ is a field.

Proof: Lowk trivial. □

Exercise 1.11. A ring A is Boolean if $x^2 = x$ for all $x \in A$. In a Boolean ring A, show that

i) $2x = 0$ for all $x \in A$;
ii) every prime ideal p is maximal, and A/p is a field with two elements;
iii) every finitely generated ideal in A is principal.

Proof: Suppose A is a Boolean ring.

i) $$x + 1 = (x+1)^2 = x^2 + 2x + 1 = x + 2x + 1 \Rightarrow 2x = 0.$$
ii) First section of the question is exercise 2.7.
iii) We want to show $(x, y) = (x + y + xy)$. One direction is trivial. The other, notice $x(x + y + xy) = x$ and $y(x + y + xy) = y$. To get the generality, proceed by induction on number of generators.

□

Definition 1.1 (Local Ring). A is a local ring if and only if there is only 1 maximal ideal $\mathfrak{m}$. $A/\mathfrak{m}$ is called the residue field of A.

Exercise 1.12. A local ring contains no idempotent $\neq 0, 1$.

Proof: Let A be a local ring. Let $x \notin \{0, 1\}$ be a idempotent of A. □

1.1 Construction of an algebraic closure of a field (E. Artin)

Exercise 1.13. Let K be a field and let Σ be the set of all irreducible monic polynomials f in one indeterminate with coefficients in K. Let A be the polynomial ring over K generated by indeterminates x_f, one for each $f \in \Sigma$. Let a be the ideal of A generated by the polynomials $f(x_f)$ for all $f \in \Sigma$. Show that $a \neq (1)$.

Let m be a maximal ideal of A containing a, and let $K_1 = A/m$. Then K_1 is an extension field of K in which each $f \in \Sigma$ has a root. Repeat the construction with K_1 in place of K, obtaining a field K_2, and so on. Let $L = \cup_{n=1}^{\infty} K_n$. Then L is a field in which each $f \in \Sigma$ splits completely

into linear factors. Let $\overline{K}$ be the set of all elements of L which are algebraic over K. Then $\overline{K}$ is an algebraic closure of K.

Exercise 1.14. In a ring A, let Σ be the set of all ideals in which every element is a zero-divisor. Show that the set Σ has maximal elements and that every maximal element of Σ is a prime ideal. Hence the set of zero-divisors in A is a union of prime ideals.

1.2 The prime spectrum of a ring

Exercise 1.15. Let A be a ring and let X be the set of all prime ideals of A. For each subset E of A, let $V(E)$ denote the set of all prime ideals of A which contain E. Prove that

i) if a is the ideal generated by E, then $V(E) = V(a) = V(r(a))$.

ii) $V(0) = X$, $V(1) = \emptyset$.

iii) if $(E_i)_{i\in I}$ is any family of subsets of A, then

$$V\left(\bigcup_{i\in I} E_i\right) = \bigcap_{i\in I} V(E_i).$$

iv) $V(a \cap b) = V(ab) = V(a) \cup V(b)$ for any ideals a, b of A.

These results show that the sets $V(E)$ satisfy the axioms for closed sets in a topological space. The resulting topology is called the **Zariski topology**. The topological space X is called the **prime spectrum** of A, and is written $\mathrm{Spec}(A)$.

Proof:

i) If $E \subseteq \mathfrak{p}$, then all finite A-linear combinations of E is contained in $\mathfrak{p}$ (by definition of ideal), thus $E \subseteq a \subseteq \mathfrak{p}$. Thus $V(E) \subseteq V(a)$. Suppose $a \subseteq \mathfrak{p} \Rightarrow E \subseteq \mathfrak{p} \Rightarrow V(a) \subseteq V(E)$. Look at

$$r(a) = \bigcap_{\substack{\mathfrak{p}\in \mathrm{Spec}(A)\\ a\subseteq\mathfrak{p}}} \mathfrak{p}.$$

Thus if $r(a) \subseteq \mathfrak{p} \Rightarrow a \subseteq \mathfrak{p} \Rightarrow V(a) \subseteq V(r(a))$. Furthermore, if $a \subseteq \mathfrak{p} \Rightarrow r(a) \subseteq \mathfrak{p} \Rightarrow V(r(a)) \subseteq V(a)$. Thus $V(a) = V(r(a))$.

ii) $V(0) := \{\mathfrak{p} \in \mathrm{Spec}(A) : (0) \subseteq \mathfrak{p}\} = \{\mathfrak{p} \in \mathrm{Spec}(A) : 0 \in \mathfrak{p}\} = \mathrm{Spec}(A)$. On the other hand $V(X) := \{\mathfrak{p} \in \mathrm{Spec}(A) : A \subseteq \mathfrak{p}\} = \emptyset$ because each prime ideal is contained in A and A itself is not prime.

iii) Let $(E_i)_{i\in I}$ be a family of subsets of A.

$$\begin{aligned} V\left(\bigcup_{i\in I} E_i\right) &= \left\{\mathfrak{p} \in \mathrm{Spec}(A) : \bigcup_{i\in I} E_i \subseteq \mathfrak{p}\right\} \\ &= \{\mathfrak{p} \in \mathrm{Spec}(A) : \text{for all } i \in I, E_i \subseteq \mathfrak{p}\} \\ &= \bigcap_{i\in I} \{\mathfrak{p} \in \mathrm{Spec}(A) : E_i \subseteq \mathfrak{p}\} \\ &= \bigcap_{i\in I} V(E_i). \end{aligned}$$

iv)
$$\begin{aligned} V(a) \cup V(b) &= \{\mathfrak{p} \in \mathrm{Spec}(A) : a \subseteq \mathfrak{p}\} \cup \{\mathfrak{p} \in \mathrm{Spec}(A) : b \subseteq \mathfrak{p}\} \\ &= \{\mathfrak{p} \in \mathrm{Spec}(A) : a \subseteq \mathfrak{p} \vee b \subseteq \mathfrak{p}\} \\ &\subseteq \{\mathfrak{p} \in \mathrm{Spec}(A) : a \cap b \subseteq \mathfrak{p}\} = V(a \cap b) \end{aligned}$$

$$V(a \cap b) = \{\mathfrak{p} \in \operatorname{Spec}(A) : a \cap b \subseteq \mathfrak{p}\} \subseteq V(ab).$$

$$V(ab) = \{\mathfrak{p} \in \operatorname{Spec}(A) : ab \subseteq \mathfrak{p}\}.$$

Suppose $ab \subset \mathfrak{p}$.Assume the contrary, that $a \not\subseteq \mathfrak{p}$ and $b \not\subseteq \mathfrak{p}$, Thus $\exists \alpha \in a - \mathfrak{p}$ and $\beta \in b - \mathfrak{p}$. $\alpha\beta = 0 \in A/\mathfrak{p} \Rightarrow \alpha \in \mathfrak{p}$ or $\beta \in \mathfrak{p}$. Thus by contradiction, $\alpha \subseteq \mathfrak{p}$ or $\beta \subseteq \mathfrak{p}$. Thus

$$V(ab) \subset V(A) \cup V(b).$$

Thus we have the chain

$$V(a) \cup V(b) \subseteq V(a \cap b) \subseteq V(ab) \subseteq V(a) \cap V(b).$$

□

Exercise 1.16. Draw pictures of $\operatorname{Spec}(\mathbb{Z})$, $\operatorname{Spec}(\mathbb{R})$, $\operatorname{Spec}(\mathbb{C}[x])$, $\operatorname{Spec}(\mathbb{R}[x])$, $\operatorname{Spec}(\mathbb{Z}[x])$.

Proof:

□

Exercise 1.17. For each $f \in A$, let X_f denote the complement of $V(f)$ in $X = \operatorname{Spec}(A)$. The sets X_f are open. Show that they form a basis of open sets for the Zariski topology, and that

i) $X_f \cap X_g = X_{fg}$;
ii) $X_f = \emptyset \Leftrightarrow f$ is nilpotent;
iii) $X_f = X \Leftrightarrow f$ is a unit;
iv) $X_f = X_g \Leftrightarrow r((f)) = r((g))$;
v) X is quasi-compact (that is, every open covering of X has a finite subcovering).
vi) More generally, each X_f is quasi-compact.
vii) An open subset of X is quasi-compact if and only if it is a finite union of sets X_f.

The sets X_f are called **basic open sets** of $X = \operatorname{Spec}(A)$.

[To prove (v), remark that it is enough to consider a covering of X by basic open sets X_{f_i} $(i \in I)$. Show that the f_i generate the unit ideal and hence that there is an equation of the form

$$1 = \sum_{i \in J} g_i f_i \quad (g_i \in A)$$

where J is some finite subset of I. Then the X_{f_i} $(i \in J)$ cover X.]

Proof:

$$\begin{aligned} X_f \cap X_g &= (V(f))^c \cap (V(g))^c \\ &= (V(f) \cup V(g))^c \\ &= (V(fg))^c = X_{fg} \end{aligned}$$

$$X_f = \emptyset \Leftrightarrow \operatorname{Spec}(A) = V(f) = \{\mathfrak{p} \in \operatorname{Spec}(A) : (f) \subseteq \mathfrak{p}\} \Leftrightarrow f \in \mathfrak{N}$$

the last iff follows from the fact that

$$\mathfrak{N} = \bigcap_{\mathfrak{p} \in \operatorname{Spec}(A)} \mathfrak{p}.$$

$$X_f = X \Leftrightarrow V(f) = \emptyset \Leftrightarrow f \text{ not in any maximal ideal} \Leftrightarrow f \text{ unit}$$

$$X_f = X_g \Leftrightarrow V(r(f)) = V(f) = V(g) = V(r(g))$$

. By definition

$$r(f) = \bigcap_{\mathfrak{p} \in V(r(f))} \mathfrak{p} = r(g).$$

Did quasi-compactness of X in homework 3 but used closed set definition of compactness. □

Exercise 1.18. For psychological reasons it is sometimes convenient to denote a prime ideal of A by a letter such as x or y when thinking of it as a point of $X = \mathrm{Spec}(A)$. When thinking of x as a prime ideal of A, we denote it by p_x (logically, of course, it is the same thing). Show that

i) the set $\{x\}$ is closed (we say that x is a "closed point") in $\mathrm{Spec}(A) \Leftrightarrow p_x$ is maximal;
ii) $\overline{\{x\}} = V(p_x)$;
iii) $y \in \overline{\{x\}} \Leftrightarrow p_x \subseteq p_y$;
iv) X is a T_0-space (this means that if x, y are distinct points of X, then either there is a neighborhood of x which does not contain y, or else there is a neighborhood of y which does not contain x).

Proof:

i) Suppose $\{x\}$ is closed in $\mathrm{Spec}(A)$. By 2. $\{x\} = V(p_x)$. Thus there are no other prime ideals (hence maximal ideals), containing p_x and hence p_x must be maximal. The opposite direction follows immediately from the definition of maximal ideals.
ii) $\overline{\{x\}} \subseteq V(p_x)$ due to $x \in V(p_x)$ and $V(p_x)$ being closed and closure is the smallest closed set containing $\{x\}$. We want to show that $V(p_x) \subseteq \overline{\{x\}}$. Let $y \in V(p_x)$, by definition $p_x \subseteq p_y$. Since X_f form a basis for the topology of Spec, we know that if a open subset $U \ni y$, then $\exists f \in p_y$ such that $X_f \subseteq U$. We want to show that $x \in X_f \Leftrightarrow x \notin V(f) \Leftrightarrow f \notin p_x$. Note that $y \in X_f \Rightarrow y \notin V(f) \Rightarrow f \notin p_y$. Since $p_x \subseteq p_y$ and if $f \notin p_y \Rightarrow f \notin p_x$. Thus $x \in X_f$.
iii) $y \in \overline{\{x\}} = V(p_x) \Rightarrow p_y \subseteq p_x$. Suppose $p_y \subseteq p_x \Rightarrow y \in V(p_x) \Rightarrow y = \overline{\{x\}}$
iv) This is obvious.

□

Exercise 1.19. A topological space X is said to be **irreducible** if $X \neq \emptyset$ and if every pair of non-empty open sets in X intersect, or equivalently if every non-empty open set is dense in X. Show that $\mathrm{Spec}(A)$ is irreducible if and only if the nilradical of A is a prime ideal.

Proof: Suppose that $\mathrm{Spec}(A)$ is irreducible, thus every nonempty open set in X intersects nontrivially or equivalently that the union of any two closed sets not equal to $\mathrm{Spec}(A)$ is also not equal to $\mathrm{Spec}(A)$. Translating this to a algebraic definition, if I, J are ideals of A which are not contained in all prime ideals of A, $I \cap J$ is a ideal of A which is not contained in all of the prime ideals of A. Note $\mathrm{Spec}(A) = V(\mathfrak{N})$. Translating this further, if $a \notin \mathfrak{N}$ and $b \notin \mathfrak{N}$ then $ab \notin \mathfrak{N}$. Meaning that if $ab \in \mathfrak{N}$ then a or b is in $\mathfrak{N}$ and $\mathfrak{N}$ is prime. Note all implications are bidirectional. □

Exercise 1.20. Let X be a topological space.

i) If Y is an irreducible (Exercise 19) subspace of X, then the closure $\overline{Y}$ of Y in X is irreducible.
ii) Every irreducible subspace of X is contained in a maximal irreducible subspace.
iii) The maximal irreducible subspaces of X are closed and cover X. They are called the **irreducible components** of X. What are the irreducible components of a Hausdorff space?
iv) If A is a ring and $X = \mathrm{Spec}(A)$, then the irreducible components of X are the closed sets $V(p)$, where p is a minimal prime ideal of A (Exercise 8).

Proof:

i) Suppose Y is an irreducible subspace of X. Let A, B be closed sets of $\overline{Y}$ such that $A \subset \overline{Y}$ and $B \subset \overline{Y}$. Note that A, B are closed in X. Note that $Y \not\subset A$ and $Y \not\subset B$ because otherwise A or B would have to be equal to $\overline{Y}$. $(Y \cap A)$ and $(Y \cap B)$ are closed subsets of Y, in the subspace topology, whose union is not Y.

$$\begin{aligned} Y &\neq (Y \cap A) \cup (Y \cap B) \\ &= Y \cap (A \cup B) \\ \Rightarrow Y &\not\subseteq A \cup B \Rightarrow A \cup B \neq \overline{Y}. \end{aligned}$$

ii) Let S be a irreducible subspace of X. $I = \{T \subseteq X : T \text{ irreducible}, T \supseteq S\}$, I is a poset with the ordering given by $\subseteq$. Let C be a chain. Consider

$$U = \bigcup_{T \in C} T,$$

we want to show that U is a irreducible subspace of X. It suffices to assume that $U = \overline{U}$, because if U is irreducible then $\overline{U}$ is irreducible and then $\overline{U} = U$ and U in chain. Let A, B be closed subsets of X where $A \subset U$ and $B \subset U$. We want to show that $A \cup B \subsetneq U$. We have a couple cases:

i) $\forall T \in S$, $T \subseteq A$ or $T \subseteq B$, then we have a contradiction.
ii) $\exists T \in S$, where $T \not\subseteq A$ and $T \not\subseteq B$. Thus $A \cap T$ and $B \cap T$ are closed under the subspace topology of T which don't both cover T. Then $T \cap (A \cup B) \neq T \Rightarrow A \cup B \neq U$.

Thus $A \cup B \neq U$. Thus U is irreducible. Thus every chain has a maximal element. Hence by Zorn's Lemma, S is contained in a maximal irreducible subspace.

iii) Closure is trivial. A singleton is irreducible and hence must lie in a maximal irreducible subspace of X.

□

Exercise 1.21. Let $\varphi : A \to B$ be a ring homomorphism. Let $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$. If $q \in Y$, then $\varphi^{-1}(q)$ is a prime ideal of A, i.e., a point of X. Hence φ induces a mapping $\varphi^* : Y \to X$. Show that

i) if $f \in A$ then $(\varphi^*)^{-1}(X_f) = Y_{\varphi^{-1}(f)}$, and hence that φ^* is continuous.
ii) if a is an ideal of A, then $\overline{(\varphi^*)^{-1}(V(a))} = V(a^e)$.
iii) if b is an ideal of B, then $\overline{\varphi^*(V(b))} = V(b^c)$.
iv) if φ is surjective, then φ^* is a homeomorphism of Y onto the closed subset $V(\ker(\varphi))$ of X. (In particular, $\mathrm{Spec}(A)$ and $\mathrm{Spec}(A/\mathfrak{N})$ (where $\mathfrak{N}$ is the nilradical of A) are naturally homeomorphic.)
v) if φ is injective, then $\varphi^*(Y)$ is dense in X. More precisely, $\varphi^*(Y)$ is dense in $X \Leftrightarrow \ker(\varphi) \subseteq \mathfrak{N}$.
vi) let $\psi : B \to C$ be another ring homomorphism. Then $(\psi \circ \varphi)^* = \varphi^* \circ \psi^*$.
vii) let A be an integral domain with just one non-zero prime ideal p, and let K be the field of fractions of A. Let $B = (A/p) \times K$. Define $\varphi : A \to B$ by $\varphi(x) = (\overline{x}, x)$, where $\overline{x}$ is the image of x in A/p. Show that φ^* is bijective but not a homeomorphism.

Proof:

i) Let $f \in A$ and $q \in Y$. Then

$$\begin{aligned} q \in (\varphi^*)^{-1}(X_f) &\Leftrightarrow \varphi^*(q) \in X_f \Leftrightarrow f \notin \varphi^{-1}(q) \\ &\Leftrightarrow \varphi(f) \notin q \Leftrightarrow q \in Y_{\varphi(f)}. \end{aligned}$$

Hence $(\varphi^*)^{-1}(X_f) = Y_{\varphi(f)}$. Since the sets X_f form a basis for the topology of X and their preimages $Y_{\varphi(f)}$ are open in Y, φ^* is continuous.

ii) $a \subseteq A$, an ideal.

$$\begin{aligned} (\varphi^*)^{-1}(V(a)) &= (\varphi^*)^{-1}(\{x \in \mathrm{Spec}(A) : a \subseteq p_x\}) \\ &= \left\{x \in \mathrm{Spec}(B) : a \subseteq p_{\varphi^*(x)}\right\} \\ &= \{x \in \mathrm{Spec}(B) : a \subseteq \varphi^{-1}(p_x)\} \\ &= \{x \in \mathrm{Spec}(B) : \varphi(a) \subseteq p_x\} \\ &= \{x \in \mathrm{Spec}(B) : (\varphi(a)) \subseteq p_x\} \\ &= \{x \in \mathrm{Spec}(B) : a^e \subseteq p_x\} \\ &= V(a^e) \end{aligned}$$

iii) $b \subseteq B$ an ideal:

$$\begin{aligned} V(b^c) &= \{x \in \mathrm{Spec}(A) : b^c \subseteq p_x\} \\ &= \{x \in \mathrm{Spec}(A) : \varphi^{-1}(b) \subseteq p_x\} \\ &= \{x \in \mathrm{Spec}(A) : b \subseteq \varphi(p_x)\} \end{aligned}$$

TODO

iv) Suppose φ is surjective. We know φ^* is continuous. Note that $B \cong A/\ker(\varphi)$ as a ring. Note the preimage of a prime ideal of B, and hence a prime ideal of $A/\ker(\varphi)$, is a prime ideal of A which contains $\ker(\varphi)$. Furthermore, note that this correspondence is bijective. Thus φ^* is a continuous bijection. We just need to show its either open or closed, I will show its closed.

$$\begin{aligned}\varphi^*(V(I)) &= \varphi^*(\{x \in \operatorname{Spec}(B) : I \subseteq p_x\}) \\ &= \{\varphi^*(x) : x \in \operatorname{Spec}(B), I \subseteq p_x\} \\ &= \{x \in V(\ker(\varphi)) : I \subseteq \varphi(p_x)\} \\ &= \{x \in V(\ker(\varphi)) : \varphi^{-1}(I) \subseteq p_x\} \\ &= V(\ker(\varphi)) \cap V(\varphi^{-1}(I))\end{aligned}$$

and thus $\varphi^*(V(I))$ is closed. Thus φ^* is a homeomorphism onto its image. Note that $I \subset \varphi(p_x) \Leftrightarrow \varphi^{-1}(I) \subseteq p_x$: Suppose $\varphi^{-1}(I) \subseteq p_x$, apply φ to both sides and because of surjectivity $\varphi(\varphi^{-1}(I)) = I$. Suppose $I \subseteq \varphi(p_x)$, $\varphi^-1(I) \subseteq \varphi^{-1}(\varphi(p_x)) = p_x + \ker(\varphi) = p_x$ because $p_x \supseteq \ker(\varphi)$.

v)
$$\begin{aligned}\overline{\varphi^*(\operatorname{Spec} B)} &= V\left(\bigcap_{x \in \operatorname{Spec}(B)} \varphi^{-1}(\mathfrak{p}_x)\right) \\ &= V\left(\varphi^{-1}\left(\bigcap_{x \in \operatorname{Spec}(B)} p_x\right)\right) \\ &= V(\varphi^{-1}(\mathfrak{N}(B))) \\ &= V\left(\sqrt{\ker \varphi}\right)\end{aligned}$$

The last equality follows, because $a \in \varphi^{-1}(\mathfrak{N}(B)) \Leftrightarrow \varphi(a)^n = 0 \Leftrightarrow \varphi(a^n) = 0 \Leftrightarrow a^n \in \ker \varphi$.

Then

$$\begin{aligned}V\left(\sqrt{\ker \varphi}\right) &= V(\ker \varphi) \\ &= \operatorname{Spec}(A)\end{aligned}$$

because $\ker \varphi \subseteq \mathfrak{N}(A)$ which is in inside all prime ideals.

vi) Functoriality

vii)

□

Exercise 1.22. Let $A = \prod_{i=1}^n A_i$ be the direct product of rings A_i. Show that $\operatorname{Spec}(A)$ is the disjoint union of open (and closed) subspaces X_i, where X_i is canonically homeomorphic with $\operatorname{Spec}(A_i)$.

Conversely, let A be any ring. Show that the following statements are equivalent:

i) $X = \operatorname{Spec}(A)$ is disconnected.
ii) $A \cong A_1 \times A_2$ where neither of the rings A_1, A_2 is the zero ring.
iii) A contains an idempotent $\neq 0, 1$.

In particular, the spectrum of a local ring is always connected (Exercise 12).

Exercise 1.23. Let A be a Boolean ring (Exercise 11), and let $X = \operatorname{Spec}(A)$.

i) For each $f \in A$, the set X_f (Exercise 17) is both open and closed in X.
ii) Let $f_1, ..., f_n \in A$. Show that $X_{f_1} \cup ... \cup X_{f_n} = X_f$ for some $f \in A$.
iii) The sets X_f are the only subsets of X which are both open and closed. [Let $Y \subseteq X$ be both open and closed. Since Y is open, it is a union of basic open sets X_f. Since Y is closed and X is quasi-compact (Exercise 17), Y is quasi-compact. Hence Y is a finite union of basic open sets; now use (ii) above.]
iv) X is a compact Hausdorff space.

Exercise 1.24. Let L be a lattice, in which the sup and inf of two elements a, b are denoted by $a \bigvee b$ and $a \bigwedge b$ respectively. L is a **Boolean lattice** (or **Boolean algebra**) if
i) L has a least element and a greatest element (denoted by $0, 1$ respectively).
ii) Each of $\bigvee, \bigwedge$ is distributive over the other.
iii) Each $a \in L$ has a unique "complement" $a' \in L$ such that $a \bigvee a' = 1$ and $a \bigwedge a' = 0$.

(For example, the set of all subsets of a set, ordered by inclusion, is a Boolean lattice.)

Let L be a Boolean lattice. Define addition and multiplication in L by the rules

$$a + b = \left(a \bigwedge b'\right) \bigvee \left(a' \bigwedge b\right), \quad ab = a \bigwedge b.$$

Verify that in this way L becomes a Boolean ring, $A(L)$.

Conversely, starting from a Boolean ring A, define an ordering on A as follows: $a \leq b$ means that $a = ab$. Show that, with respect to this ordering, A is a Boolean lattice. [The sup and inf are given by $a \bigvee b = a + b + ab$ and $a \bigwedge b = ab$, and the complement by $a' = 1 - a$.] In this way we obtain a one-to-one correspondence between (isomorphism classes of) Boolean rings and (isomorphism classes of) Boolean lattices.

Exercise 1.25. From the last two exercises deduce Stone's theorem, that every Boolean lattice is isomorphic to the lattice of open-and-closed subsets of some compact Hausdorff topological space.

Exercise 1.26. Let A be a ring. The subspace of $\mathrm{Spec}(A)$ consisting of the maximal ideals of A, with the induced topology, is called the **maximal spectrum** of A and is denoted by $\mathrm{Max}(A)$. For arbitrary commutative rings it does not have the nice functorial properties of $\mathrm{Spec}(A)$ (see Exercise 21), because the inverse image of a maximal ideal under a ring homomorphism need not be maximal.

Let X be a compact Hausdorff space and let $C(X)$ denote the ring of all real-valued continuous functions on X (add and multiply functions by adding and multiplying their values). For each $x \in X$, let m_x be the set of all $f \in C(X)$ such that $f(x) = 0$. The ideal m_x is maximal, because it is the kernel of the (surjective) homomorphism $C(X) \to \mathbb{R}$ which takes f to $f(x)$. If $\tilde{X}$ denotes $\mathrm{Max}(C(X))$, we have therefore defined a mapping $\mu : X \to \tilde{X}$, namely $x \mapsto m_x$.

We shall show that μ is a homeomorphism of X onto $\tilde{X}$.

i) Let m be any maximal ideal of $C(X)$, and let $V = V(m)$ be the set of common zeros of the functions in m: that is,

$$V = \{x \in X : f(x) = 0 \text{ for all } f \in m\}.$$

Suppose that V is empty. Then for each $x \in X$ there exists $f_x \in C(X)$ such that $f_x(x) \neq 0$. Since f_x is continuous, there is an open neighborhood U_x of x in X on which f_x does not vanish. By compactness a finite number of the neighborhoods, say $U_{x_1}, ..., U_{x_n}$, cover X. Let

$$f = f_{x_1}^2 + ... + f_{x_n}^2.$$

Then f does not vanish at any point of X, hence is a unit in $C(X)$. But this contradicts $f \in m$, hence V is not empty.

Let x be a point of V. Then $m \subseteq m_x$, hence $m = m_x$ because m is maximal. Hence μ is surjective.

ii) By Urysohn's lemma (this is the only non-trivial fact required in the argument) the continuous functions separate the points of X. Hence $x \neq y \Rightarrow m_x \neq m_y$, and therefore μ is injective.

iii) Let $f \in C(X)$; let

$$U_f = \{x \in X : f(x) \neq 0\}$$

and let

$$\tilde{U}_f = \{m \in \tilde{X} : f \nmid m\}.$$

Show that $\mu(U_f) = \tilde{U}_f$. The open sets U_f (resp. $\tilde{U}_f$) form a basis of the topology of X (resp. $\tilde{X}$) and therefore μ is a homeomorphism.

Thus X can be reconstructed from the ring of functions $C(X)$.

1.3 Affine algebraic varieties

Exercise 1.27. Let k be an algebraically closed field and let

$$f_\alpha(t_1, ..., t_n) = 0$$

be a set of polynomial equations in n variables with coefficients in k. The set X of all points $x = (x_1, ..., x_n) \in k^n$ which satisfy these equations is an **affine algebraic variety**.

Consider the set of all polynomials $g \in k[t_1, ..., t_n]$ with the property that $g(x) = 0$ for all $x \in X$. This set is an ideal $I(X)$ in the polynomial ring, and is called the **ideal of the variety** X. The quotient ring

$$P(X) = k[t_1, ..., t_n]/I(X)$$

is the ring of polynomial functions on X, because two polynomials g, h define the same polynomial function on X if and only if $g - h$ vanishes at every point of X, that is, if and only if $g - h \in I(X)$.

Let ξ_i be the image of t_i in $P(X)$. The ξ_i $(1 \leq i \leq n)$ are the **coordinate functions** on X: if $x \in X$, then $\xi_i(x)$ is the ith coordinate of x. $P(X)$ is generated as a k-algebra by the coordinate functions, and is called the **coordinate ring** (or **affine algebra**) of X.

As in Exercise 26, for each $x \in X$ let m_x be the ideal of all $f \in P(X)$ such that $f(x) = 0$; it is a maximal ideal of $P(X)$. Hence, if $\tilde{X} = \mathrm{Max}(P(X))$, we have defined a mapping $\mu : X \to \tilde{X}$, namely $x \mapsto m_x$.

It is easy to show that μ is injective: if $x \neq y$, we must have $x_i \neq y_i$ for some i $(1 \leq i \leq n)$, and hence $\xi_i - y_i$ is in m_x but not in m_y, so that $m_x \neq m_y$. What is less obvious (but still true) is that μ is surjective. This is one form of Hilbert's Nullstellensatz (see Chapter 7).

Exercise 1.28. Let $f_1, ..., f_m$ be elements of $k[t_1, ..., t_n]$. They determine a **polynomial mapping** $\varphi : k^n \to k^m$: if $x \in k^n$, the coordinates of $\varphi(x)$ are $f_1(x), ..., f_{m(x)}$. Let X, Y be affine algebraic varieties in k^n, k^m respectively. A mapping $\varphi : X \to Y$ is said to be **regular** if φ is the restriction to X of a polynomial mapping from k^n to k^m.

If η is a polynomial function on Y, then $\eta \circ \varphi$ is a polynomial function on X. Hence φ induces a k-algebra homomorphism $P(Y) \to P(X)$, namely $\eta \mapsto \eta \circ \varphi$. Show that in this way we obtain a one-to-one correspondence between the regular mappings $X \to Y$ and the k-algebra homomorphisms $P(Y) \to P(X)$.